

ACML100

Lecture 7

Guggenheim Method

$$\ln \left(\frac{P_t - P_\infty}{P_0 - P_\infty} \right) = -kt$$

$$\left(\frac{P_t - P_\infty}{P_0 - P_\infty} \right) = \exp(-kt)$$

$$(P_t - P_\infty) = (P_0 - P_\infty) \exp(-kt)$$

$$P_t = (P_0 - P_\infty) \exp(-kt) + P_\infty$$

$$P_{t+\tau} = (P_0 - P_\infty) \exp[-k(t + \tau)] + P_\infty$$

$$P_t - P_{t+\tau} = (P_0 - P_\infty)(1 - \exp(-k\tau)) \exp(-kt)$$

$$\ln(P_t - P_{t+\tau}) = \ln(P_0 - P_\infty) + \ln(1 - \exp(k\tau)) - kt$$

Kezdy Swinbourne Method

$$(P_t - P_\infty) = (P_0 - P_\infty) \exp(-kt)$$

$$P_{t+\tau} - P_\infty = (P_0 - P_\infty) \exp[-k(t + \tau)]$$

$$\frac{(P_t - P_\infty)}{(P_{t+\tau} - P_\infty)} = \exp(kt)$$

$$(P_t - P_\infty) = (P_{t+\tau} - P_\infty) \exp(kt)$$

$$P_t = (1 - \exp(kt))P_\infty + P_{t+\tau} \exp(kt)$$

Reaction order and reaction mechanism

Most important application of studying the order of a reaction is to establish the mechanism of reaction.

Reactions proceed in one or more elementary steps.

Molecularity: The number of molecules that come together to react in one elementary step

In elementary step
molecularity = order of reaction

$A \rightarrow \text{products}$	1 st order	rate = $k[A]$	Unimolecular
$A \rightarrow \text{products}$	2 nd order	rate = $k[A]^2$	Bimolecular

Overall order of reaction corresponds to stoichiometric equation => The reaction mechanism most probably involves one elementary step that is identical to stoichiometric equation.

When the reaction order does not corresponds to the stoichiometry of the reaction, the reaction certainly involves more than one elementary reaction.

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Elementary Reaction

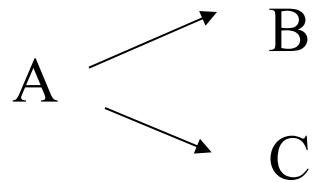
(i) Reversible elementary reaction



(ii) Consecutive elementary reaction



(iii) Parallel Reaction:



Consecutive Reaction



$$\frac{d[A]}{dt} = -k_a[A] \quad [A] = [A]_0 \exp(-k_a t)$$

$$\frac{d[I]}{dt} = k_a[A] - k_b[I]$$

$$\frac{d[I]}{dt} = k_a[A]_0 \exp(-k_a t) - k_b[I]$$

Consecutive Reaction

$$\frac{d[I]}{dt} = k_a[A]_0 \exp(-k_a t) - k_b[I]$$

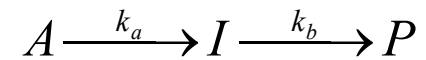
$$\frac{d[I]}{dt} + k_b[I] = k_a[A]_0 \exp(-k_a t)$$

$$[I] = \frac{k_a}{k_a - k_b} [\exp(-k_a t) - \exp(-k_b t)] [A]_0$$

$$\begin{aligned} [P] &= [A]_0 - [A] - [I] \\ &= [A]_0 \left\{ 1 + \frac{k_a \exp(-k_b t) - k_b \exp(-k_a t)}{k_b - k_a} \right\} \end{aligned}$$

Consecutive Reaction

$$[P] = [A]_0 \left\{ 1 + \frac{k_a \exp(-k_b t) - k_b \exp(-k_a t)}{k_b - k_a} \right\}$$



$$\text{if } k_b \gg k_a$$

$$\exp(-k_b t) \ll \exp(-k_a t)$$

$$\text{Also } k_b - k_a \approx k_b$$

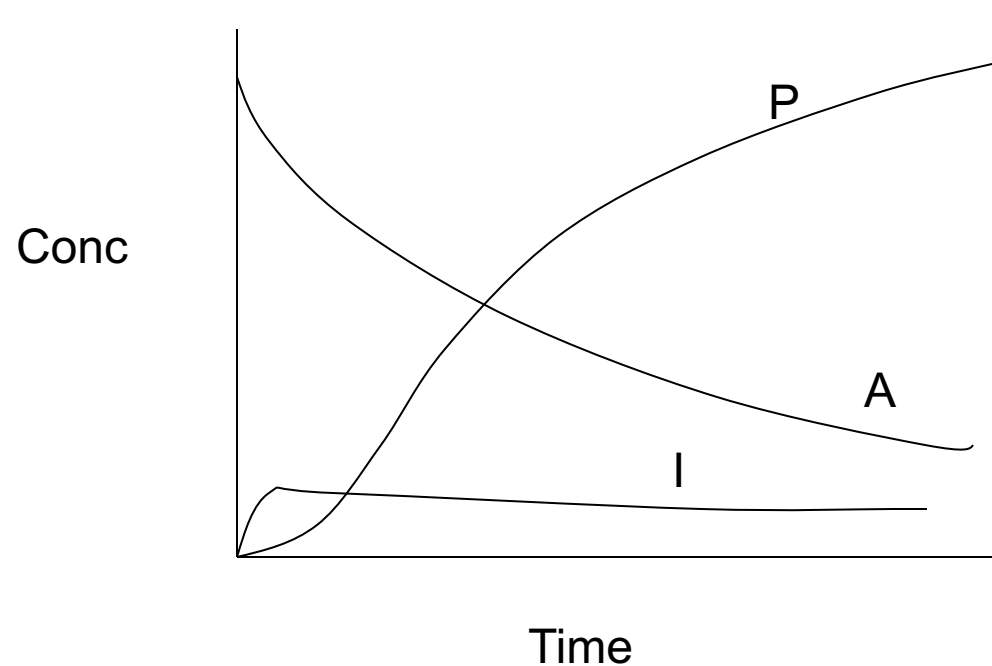
$$[P] = (1 - \exp(-k_a t)) [A]_0$$

Consecutive Reaction

- Rate of formation of the final product P depends on only the smaller of two rate constants.



Consecutive Reaction



This is the basis
of steady-State
approximation.

$$\frac{d[I]}{dt} \approx 0$$

Consecutive Reaction



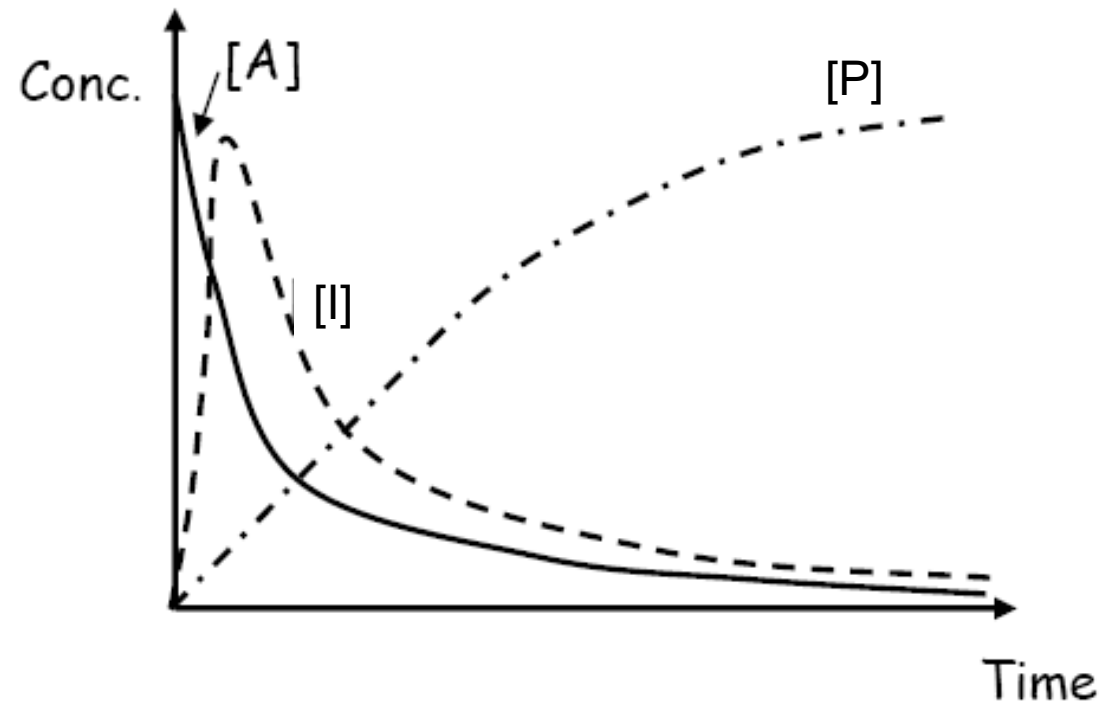
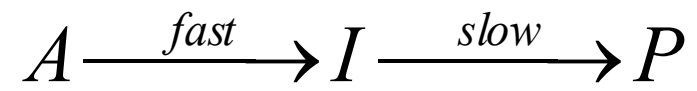
$$\frac{d[I]}{dt} = k_a [A] - k_b [I] \approx 0$$

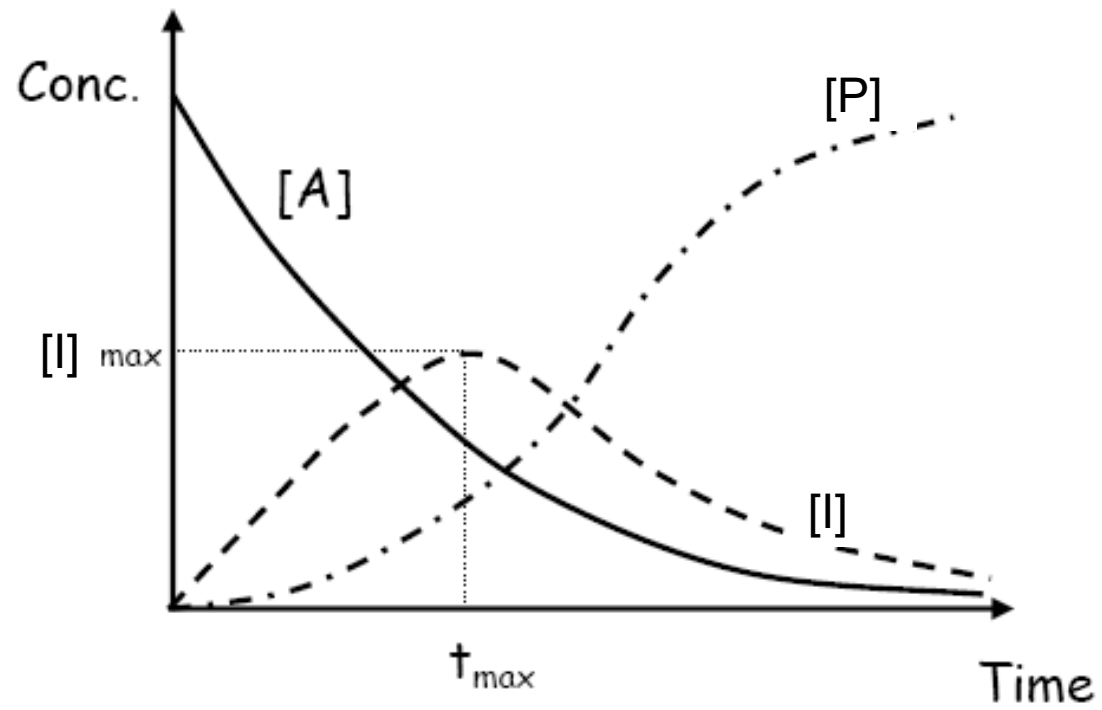
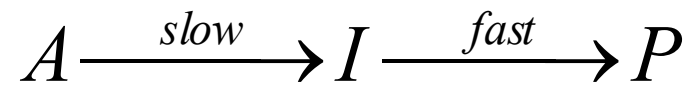
$$[I] = \frac{k_a}{k_b} [A]$$

$$\begin{aligned} \frac{d[P]}{dt} &= k_b [I] \approx k_a [A] \\ &= k_a [A]_0 \exp(-k_a t) \end{aligned}$$

$$P = [1 - \exp(-k_a t)] [A]_0$$

Case II: when $k_2 \gg k_1$





$$t_{\max}^B = \frac{\ln(k_1 / k_2)}{k_1 - k_2} \quad \text{with} \quad k_1 \neq k_2 \quad [I]_{\max} = \frac{k_1}{k_2} [A]_0 e^{-k_1 t_{\max}^B}$$

When are consecutive and Single-Step reactions distinguishable ?

$$k_1 \gg k_2$$

$$[A] = [A]_0 e^{-k_1 t}$$

$$[P] = (1 - \exp(-k_2 t)) [A]_0$$

Rate of formation of product is different from rate of decay of A

$$k_1 \ll k_2$$

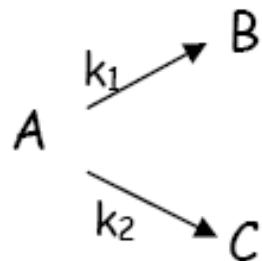
$$[A] = [A]_0 e^{-k_1 t}$$

$$[P] = (1 - \exp(-k_1 t)) [A]_0$$

Rate of formation of product is same as rate of decay of A

Parallel Reaction

Parallel 1st order reactions



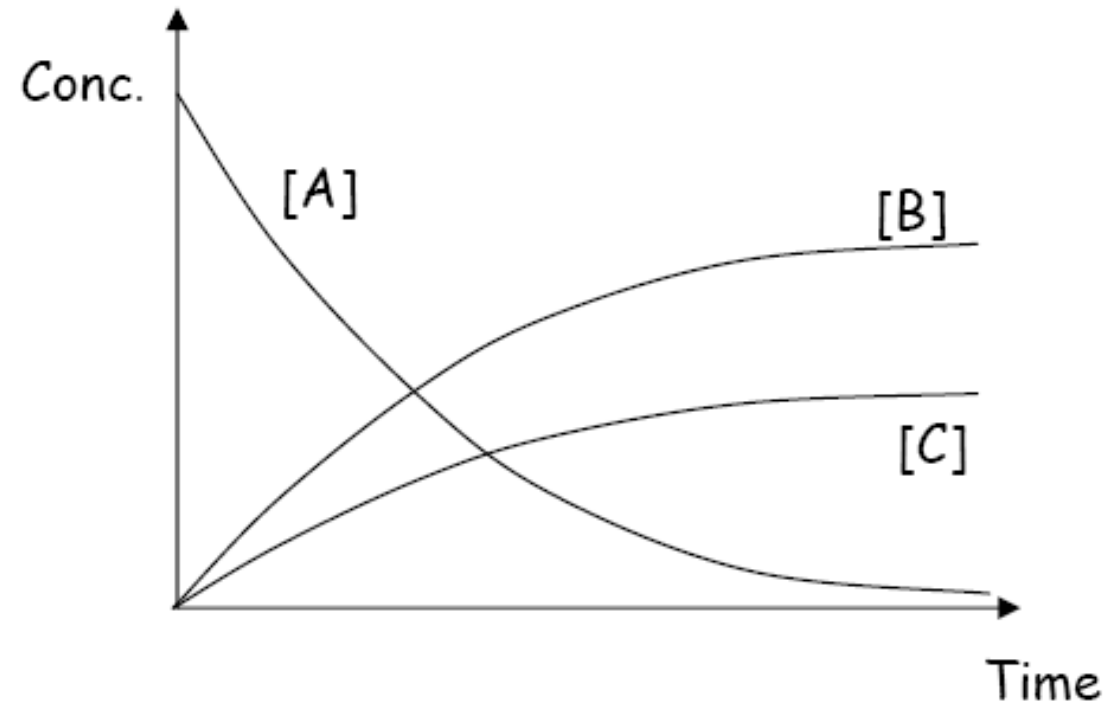
$$-\frac{d[A]}{dt} = k_1[A] + k_2[A]$$

$$[A] = [A]_0 e^{-(k_1+k_2)t}$$

$$[B] = \frac{k_1[A]_0}{k_1 + k_2} (1 - e^{-(k_1+k_2)t})$$

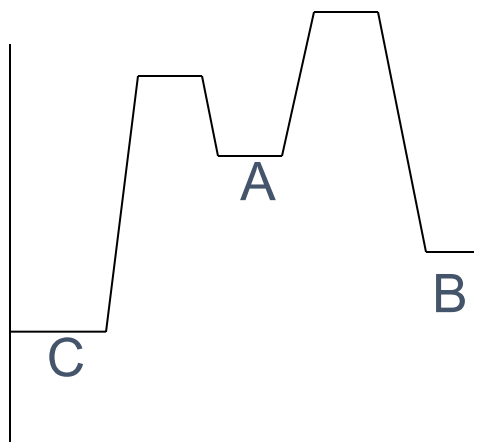
$$[C] = \frac{k_2[A]_0}{k_1 + k_2} (1 - e^{-(k_1+k_2)t})$$

Parallel Reaction

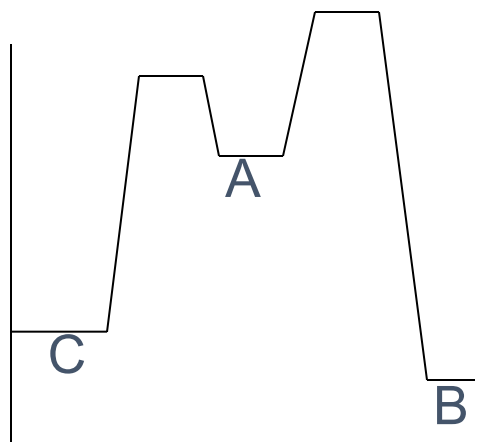
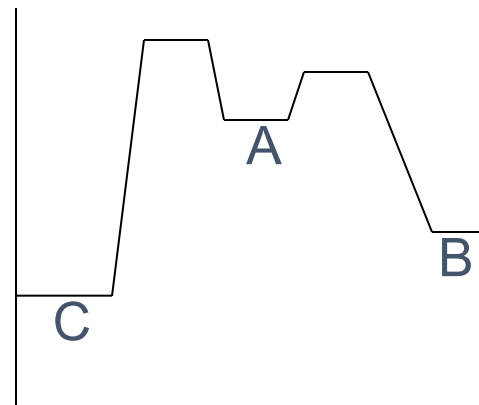


Thermodynamic vs. Kinetic Control

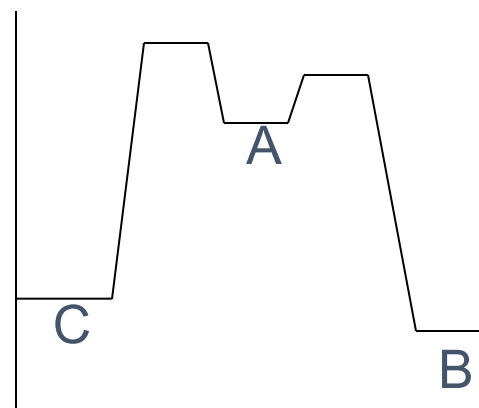
$$k_2 > k_1, K_2 > K_1$$



$$k_2 < k_1, K_2 > K_1$$

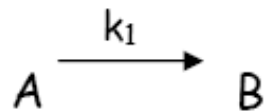


$$k_2 > k_1, K_2 < K_1$$

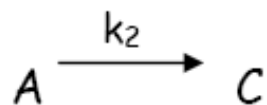


$$k_2 < k_1, K_2 < K_1$$

Parallel Reaction



$$\frac{d[B]}{dt} = k_1[A]$$



$$\frac{d[C]}{dt} = k_2[A]^2$$

$$-\frac{d[A]}{dt} = k_1[A] + k_2[A]^2$$

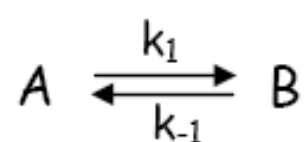
$$[A] = \frac{k_1[A]_0}{e^{k_1 t} (k_1 + k_2[A]_0) - k_2[A]_0}$$

Limiting cases:

$$\text{i) } k_2[A]_0 \ll k_1 \quad \Rightarrow \quad [A] = [A]_0 e^{-k_1 t}$$

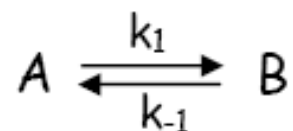
$$\text{ii) } k_2[A]_0 \gg k_1 \quad \Rightarrow \quad \frac{1}{[A]} = \frac{1}{[A]_0} + k_2 t$$

Reversible Reaction



$$K_{eq} = \frac{[B]_{eq}}{[A]_{eq}}$$

1st order reversible reactions



$$-\frac{d[A]}{dt} = k_1[A] - k_{-1}[B]$$

$$[B] = [B]_0 + ([A]_0 - [A])$$

$$\text{So... } -\frac{d[A]}{dt} = k_1[A] - k_{-1}([B]_0 + [A]_0 - [A])$$

Reversible reaction

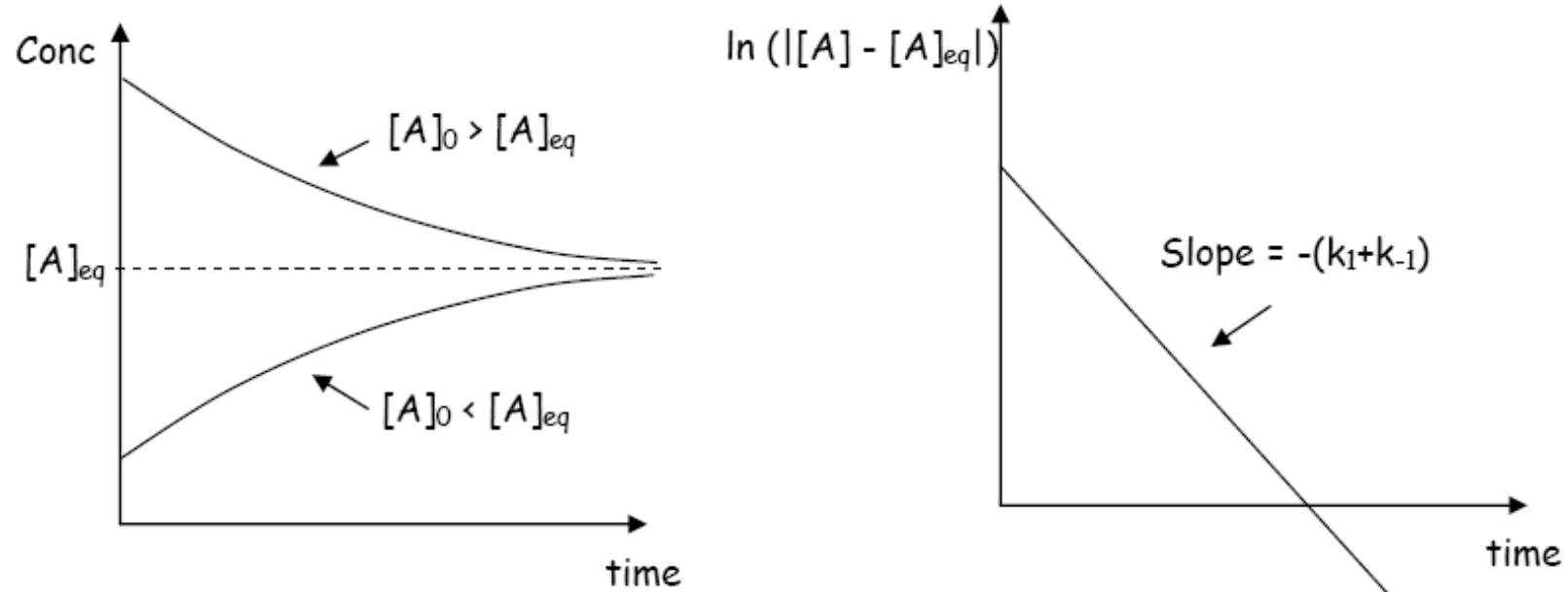
$$\text{At Equilibrium, } \frac{d[A]}{dt} = 0$$

$$\Rightarrow [A]_{eq} = \frac{k_{-1}}{k_1 + k_{-1}} ([B]_0 + [A]_0)$$

$$-\frac{d([A] - [A]_{eq})}{dt} = -\frac{d([A])}{dt} = (k_1 + k_{-1})([A] - [A]_{eq})$$

$$\Rightarrow [A] - [A]_{eq} = ([A]_0 - [A]_{eq}) e^{-(k_1 + k_{-1})t}$$

Reversible reaction



Can measure: $K_{eq} = \frac{k_1}{k_{-1}}$ and $k_1 + k_{-1} \equiv k_{obs}$

And extract k_1 and k_{-1}